

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

WINTER SEMESTER, 2023-2024
FULL MARKS: 75

CSE 4305: Computer Organization and Architecture

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. In general-purpose computers, multiple tasks are performed by a Real-Time Operating System (RTOS). As multiple tasks occur simultaneously, there has to be a task handling mechanism implemented in the system. Also, there are different mechanisms implemented to handle I/O operations in the system.

- a) Design an appropriate interrupt mechanism and show the execution sequence by the RTOS for the tasks in Table 1. Show the timestamps. Here, the largest value in the Priority column denotes the highest priority.

13
(CO4)
(PO2)

Table 1: Data for Question 1.a

Task	Execution Time(s)	Arrival Time, t (s)	Priority
Network Packet Handling	20	0	4
User Input Handling	8	10	2
Sensor Data Processing	10	5	3
System Log Update	2	12	5
Backup Operation	15	18	1

- b) In a particular system, the processor scans the keyboard every 300 milliseconds. On average, the keyboard sends 30 commands during a 3-hour period.

- i. How many of the processor's scans are unnecessary (i.e., no command is entered) during this 3-hour span?
- ii. What technique could the processor implement to enhance efficiency and reduce unnecessary scans in this scenario? Explain how the technique works with an appropriate flowchart.

5
(CO2)
(PO2)

12
(CO4)
(PO3)

2. Elexi Corp specializes in running complex Machine Learning (ML) models that require extensive floating-point computations. As demand for their services grows, they have decided to develop a new and more efficient system to enhance the performance of their ML model training tasks. This new system will take some time to complete as they are integrating several key improvements. The improvements include the addition of built-in functions that may be necessary for future ML models, the removal of dependencies among certain functionalities to enable multitasking, and the incorporation of a custom accelerator chip to improve matrix operations which are heavily used in ML computations. A certain task 'Meta' takes 720 milliseconds to execute. In that task, matrix operations account for 70% of the time, the similarity calculations take 25% of the time, and other operations take the remaining 5%. Improving the matrix operation makes the task 'Meta' 2 times faster.

- a) i. What architectural design ideas were applied in the development of the new system? Explain briefly how these ideas contribute to improving system performance.

6
(CO1)
(PO1)

ii. What additional architectural design ideas could have been incorporated into the new system? Explain your reasoning.

4
(CO1)
(PO1)

b) Based on the given scenario, with proper mathematical reasonings, derive how much the new custom accelerator chip has improved the speed of the matrix operation.

10
(CO2)
(PO2)

3. Consider the Code Snippet 1 to answer the following questions,

```

1 int main()
2 {
3     int numbers[] = {10, 20, 30, 40, 50};
4     int s = 5
5     int c = 4
6     int i = 0;
7     int sum = 0;
8     int diff = numbers[4];
9
10    while (i < size) {
11        sum += c;
12        if (i == c) {
13            numbers[1] = diff;
14        }
15        i++;
16    }

```

Code Snippet 1: A partial C program for Question 3

a) In a 32-bit system, the C program is executed. The variables *s*, *c*, *i*, *sum*, and *diff* correspond to the registers *x5*, *x7*, *x15*, *x16*, and *x17*, respectively. The base address of the array *numbers[]* is stored in register *x25*.

Table 2: Data for Question 3.a.

funct7	funct3	Opcode	Type
0000000	000	0110011	ADD
0100000	000	0110011	SUB
0000000	100	0110011	XOR
-	000	0010011	ADDI
-	010	0010011	SLTI
-	010	0000011	LW
-	010	0100011	SW
-	000	1100011	BEQ
-	001	1100011	BNE
-	100	1100011	BLT
-	101	1100011	BGE

i. Compile the C program from line 8 to 15 into corresponding RISC-V assembly instructions.

10
(CO3)
(PO2)

ii. Convert the RISC-V instructions generated for lines 8, 12, 13, and 15 into machine codes using the information provided in Table 2.

10
(CO3)
(PO2)

b) Define stored-program concept.

5
(CO1)
(PO1)